

WEEK 2 TECHNICAL REPORT

Deep Learning: Neural Networks, CNNs and Hyperparameter Experimentation

1. Objective

The objective of Week 2 was to implement and understand neural-network based learning using PyTorch. The work covered a feed-forward neural network, a Convolutional Neural Network (CNN) for image classification, and controlled hyperparameter experiments.

2. Feed-forward Neural Network

A feed-forward neural network was implemented using PyTorch for binary classification. The model used fully connected layers with ReLU activations and a sigmoid output. Binary Cross-Entropy loss and the Adam optimizer were used. After feature normalization and further training, the model achieved 100% accuracy on the given dataset.

3. CNN Image Classifier

A CNN was implemented using the CIFAR-10 dataset containing 50,000 training images and 10,000 test images across 10 classes. The model used two convolutional layers, ReLU activation, max-pooling and fully connected layers. Cross-Entropy loss and Adam with a learning rate of 0.001 were used. After 5 epochs, the model achieved 72.22% test accuracy.

4. Hyperparameter Experimentation

Experiment	LR	Batch	Epochs	Final Loss	Accuracy
Baseline	0.001	64	3	0.7991	69.40%
Higher LR	0.01	64	3	2.3036	10.00%
Larger Batch	0.001	128	3	0.8796	67.55%

The baseline configuration achieved the highest accuracy. Increasing the learning rate to 0.01 caused performance to fall to 10%, indicating that the higher learning rate was unsuitable for this configuration. Increasing the batch size to 128 produced 67.55% accuracy, slightly below the baseline.

5. Key Observations

- Training loss decreased across the successful experiments.
- The CNN achieved 72.22% test accuracy after 5 epochs.
- Hyperparameters had a significant effect on model performance.
- Learning rate 0.001 with batch size 64 gave the best result among the tested configurations.

6. Conclusion

Week 2 provided practical experience with PyTorch-based deep learning. The experiments demonstrated model training, loss calculation, backpropagation, CNN-based image classification, and hyperparameter evaluation. The results showed that suitable learning-rate and batch-size selection is important for achieving good model performance.

Tools: Python, PyTorch, Torchvision, CIFAR-10, Google Colab